UNIT 2 TEST B: WRITTEN

UNIT 2 TEST · DAY 2 · WRITTEN · VERSION B · TEACHER KEY

NAME DATE PERIOD SCORE / 14

You may use a periodic table. Write every configuration in order of filling. In the orbital diagram, each electron gets one arrow.

Part 1 Longhand configurations

8 points

Number 1 also needs its orbital diagram and the number of half-filled orbitals.

1 — 4 pts

14 **Si** Silicon 28.09

Configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$ **Half-filled orbitals** 2

Orbital diagram



2 — 2 pts

34 **Se** Selenium 78.97

Configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^4$

3 — 2 pts

22 **Ti** Titanium 47.87

Configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$

1 is 4 pts, configuration 2, diagram 1, half-filled count 1. 2 and 3 are 2 pts each, 1 off for one wrong sublevel.

Part 2 Noble-gas shorthand

6 points

4

74 **W** Tungsten 183.84

$[Xe]\ 6s^2\ 4f^{14}\ 5d^4$

5

29 **Cu** Copper 63.55

$[Ar]\ 4s^1\ 3d^{10}$

6

39 **Y** Yttrium 88.91

$[Kr]\ 5s^2\ 4d^1$

2 pts each, noble-gas core 1 and the rest in order 1. Copper is the exception, so the predicted $[Ar]\ 4s^2\ 3d^9$ earns 1.